Measure Theoretic Probability
Lesson 10
Independence of Classes, π-Systems, λ-Systems and Dynkin's Theorem

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Outline

1. Recap and the Central Question
2. Independence of Classes
3. Failure Without Extra Structure
4. π-Systems and λ-Systems
5. Dynkin's π-λ Theorem
6. Independence Survives Generating σ-Fields
7. Summary and Exercises

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Recap: Independence of Events

Two events

$A, B \in \mathcal{F}$ are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$.

Finitely many events

$A_1, \ldots, A_n$ are **mutually independent** if the probability of the intersection factors for the full n-fold intersection, and also for every sub-collection of $n-1,\ n-2, \ldots, 2$ of the sets (all pairs, triples, $\ldots$).

Infinitely many events (countable or uncountable)

A collection is independent if **every finite sub-collection** is mutually independent.

The Central Question

Today's step

Move from **sets** within one collection being independent to **whole collections of sets** being independent of each other.

Question

Suppose $\mathcal{A}_1, \ldots, \mathcal{A}_n$ are independent classes. If we enlarge each class to the σ-field $\sigma(\mathcal{A}_i)$ it generates, are the σ-fields still independent?

Answer in advance

- In general: **No**.
- With one extra, mild assumption (each $\mathcal{A}_i$ is a π-system): **Yes**.

The tool that makes this work is **Dynkin's π-λ theorem**.

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Definition: Independent Classes

Definition

Let $\mathcal{A}_1, \ldots, \mathcal{A}_n \subseteq \mathcal{F}$ be classes of sets (they need not be fields or σ-fields). They are **independent** if

$$\mathbb{P}(A_{k_1} \cap \cdots \cap A_{k_J}) = \prod_{i=1}^{J} \mathbb{P}(A_{k_i})$$

for every choice of distinct indices $k_1 < \cdots < k_J$ and every choice $A_{k_i} \in \mathcal{A}_{k_i}$.

A Cleaner Way: Allow Ω as a Choice

Equivalent definition

$\mathcal{A}_1, \ldots, \mathcal{A}_n$ are independent if

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \prod_{i=1}^{n} \mathbb{P}(A_i) \quad \text{whenever each } A_i \in \mathcal{A}_i \ \textbf{or} \ A_i = \Omega.$$

Why this works

Choosing $A_2 = \Omega$ turns $\mathbb{P}(A_1 \cap A_2 \cap A_3)$ into $\mathbb{P}(A_1 \cap A_3)$, because $\mathbb{P}(\Omega) = 1$. So allowing Ω captures all the lower-order intersections in a single statement.

Notation

Put $\mathcal{B}_i = \mathcal{A}_i \cup \{\Omega\}$. Then $\mathcal{A}_1, \ldots, \mathcal{A}_n$ are independent iff $\mathbb{P}(B_1 \cap \cdots \cap B_n) = \prod_i \mathbb{P}(B_i)$ for all $B_i \in \mathcal{B}_i$.

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A Counterexample: Two Fair Coin Tosses

Setting

$\Omega = \{HH, HT, TH, TT\}$ with each outcome of probability $\frac{1}{4}$. Define

$$A = \{HH, HT\}, \quad B = \{HH, TT\}, \quad C = \{HT, TT\},$$

$$\mathcal{A}_1 = \{A, B\}, \qquad \mathcal{A}_2 = \{C\}.$$

$\mathcal{A}_1$ and $\mathcal{A}_2$ are independent

Two checks: $A \cap C = \{HT\}$ and $B \cap C = \{TT\}$. Each has probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = \mathbb{P}(A)\mathbb{P}(C) = \mathbb{P}(B)\mathbb{P}(C)$.

The Counterexample: Generated σ-Fields Are Not Independent

$C \in \sigma(\mathcal{A}_1)$

$$(A \cup B) \setminus (A \cap B) = \{HH, HT, TT\} \setminus \{HH\} = \{HT, TT\} = C.$$

Since $\sigma(\mathcal{A}_1)$ is closed under unions, intersections and complements, $C \in \sigma(\mathcal{A}_1)$. Also $C \in \mathcal{A}_2 \subseteq \sigma(\mathcal{A}_2)$.

C is not independent of itself

C would be independent of itself iff $\mathbb{P}(C) = \mathbb{P}(C)^2$, i.e. $\mathbb{P}(C) \in \{0, 1\}$. But $\mathbb{P}(C) = \frac{1}{2}$ and $\frac{1}{2} \neq \frac{1}{4}$.

Conclusion

C lies in both $\sigma(\mathcal{A}_1)$ and $\sigma(\mathcal{A}_2)$, so these σ-fields are **not** independent, although $\mathcal{A}_1$ and $\mathcal{A}_2$ were. Note that $\mathcal{A}_1$ is not closed under intersection: $A \cap B = \{HH\} \notin \mathcal{A}_1$.

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Definition: π-System

Definition

A class $\mathcal{P}$ of subsets of Ω is a π-**system** if it is closed under **finite intersections**:
$A, B \in \mathcal{P} \Rightarrow A \cap B \in \mathcal{P}$.

Remarks

- A weak condition: every field and every σ-field is a π-system.
- The converse fails: a π-system need not contain Ω or be closed under complements.
- Example: $\{(-\infty, a] : a \in \mathbb{R}\}$ is a π-system, but not a field.

Definition: λ-System

Definition

A class $\mathcal{L}$ of subsets of Ω is a λ-**system** if

(i) $\Omega \in \mathcal{L}$;

(ii) $A \in \mathcal{L} \Rightarrow A^c \in \mathcal{L}$;

(iii) $A_1, A_2, \cdots \in \mathcal{L}$ **pairwise disjoint** $\Rightarrow \bigcup_n A_n \in \mathcal{L}$.

Comparison with a σ-field

(i) and (ii) are identical to the σ-field axioms. In (iii) the union only needs to be closed for **disjoint** sequences. Hence every σ-field is a λ-system, but a λ-system need not be a σ-field.

A λ-system that is not a σ-field

$\Omega = \{1, 2, 3, 4\}$, $\mathcal{L} = \{\text{sets of even cardinality}\}$. It satisfies (i)–(iii), but $\{1, 2\}, \{2, 3\} \in \mathcal{L}$ while $\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\} \notin \mathcal{L}$.

Equivalent Axiom: Replacing (ii) by (ii′)

Claim

(i), (ii), (iii) $\iff$ (i), (ii′), (iii), where

(ii′) $A, B \in \mathcal{L}$ and $A \subseteq B \;\Rightarrow\; B \setminus A \in \mathcal{L}$.

(i), (ii′) $\Rightarrow$ (ii)

Take $B = \Omega \in \mathcal{L}$. For $A \in \mathcal{L}$ we have $A \subseteq \Omega$, so $\Omega \setminus A = A^c \in \mathcal{L}$.

(i), (ii), (iii) $\Rightarrow$ (ii′)

Let $A \subseteq B$ in $\mathcal{L}$. Then $B^c \in \mathcal{L}$ by (ii), and A, B^c are **disjoint**, so $A \cup B^c \in \mathcal{L}$ by (iii). By (ii) again and De Morgan,

$$B \setminus A = B \cap A^c = \left(A \cup B^c\right)^c \in \mathcal{L}. \qquad \square$$

Lemma: π-System + λ-System = σ-Field

Lemma

If $\mathcal{A}$ is both a π-system and a λ-system, then $\mathcal{A}$ is a σ-field.

Proof

$\Omega \in \mathcal{A}$ and closure under complements are exactly (i) and (ii) of a λ-system. It remains to show closure under countable unions. Let $A_1, A_2, \cdots \in \mathcal{A}$ and put

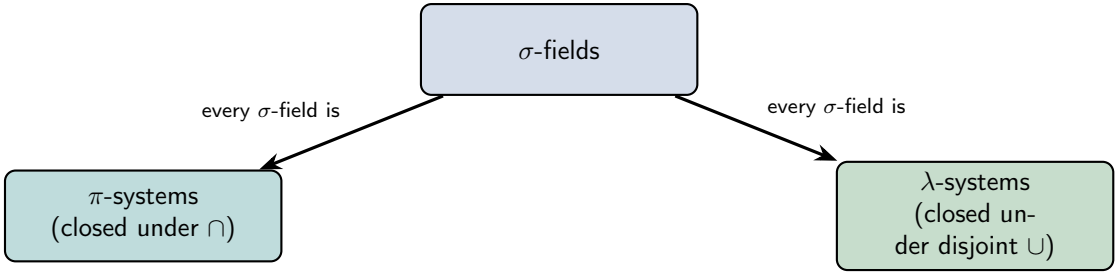
$$B_1 = A_1, \qquad B_n = A_n \cap A_1^c \cap \cdots \cap A_{n-1}^c \quad (n \geq 2).$$

Each $B_n \in \mathcal{A}$ (complements by (ii), finite intersections since $\mathcal{A}$ is a π-system). The B_n are pairwise disjoint and $\bigcup_n B_n = \bigcup_n A_n$. By (iii), $\bigcup_n A_n \in \mathcal{A}$. □

Idea

A non-disjoint union is rewritten as a disjoint union; the π-property is what keeps the pieces B_n inside the class.

How the Classes Relate



π-system $\cap$ λ-system $=$ σ-field

Lemma: Intersection of λ-Systems

Lemma

If $\mathcal{L}_1$ and $\mathcal{L}_2$ are λ-systems, then $\mathcal{L}_1 \cap \mathcal{L}_2$ (the sets that belong to **both**) is a λ-system. The same holds for any family of λ-systems.

Proof

(i) $\Omega \in \mathcal{L}_1$ and $\Omega \in \mathcal{L}_2$, so $\Omega \in \mathcal{L}_1 \cap \mathcal{L}_2$.

(ii) If $A \in \mathcal{L}_1 \cap \mathcal{L}_2$, then $A^c \in \mathcal{L}_1$ and $A^c \in \mathcal{L}_2$.

(iii) If $A_1, A_2, \cdots \in \mathcal{L}_1 \cap \mathcal{L}_2$ are disjoint, their union is in $\mathcal{L}_1$ and in $\mathcal{L}_2$.

Consequence

For any class $\mathcal{P}$ there is a **smallest λ-system containing** $\mathcal{P}$: the intersection of all λ-systems containing $\mathcal{P}$ (the power set is one, so the family is non-empty).

Lemma: The Class $\mathcal{L}_A$

Lemma

Let $\mathcal{L}$ be a λ-system and $A \in \mathcal{L}$. Define

$$\mathcal{L}_A = \{B \subseteq \Omega : \ A \cap B \in \mathcal{L}\}.$$

Then $\mathcal{L}_A$ is a λ-system.

Proof

(i) $A \cap \Omega = A \in \mathcal{L}$, so $\Omega \in \mathcal{L}_A$.

(ii) Let $B \in \mathcal{L}_A$, so $A \cap B \in \mathcal{L}$. Now $A \cap B \subseteq A$ with both in $\mathcal{L}$, so by (ii′)

$$A \setminus (A \cap B) = A \cap (A^c \cup B^c) = A \cap B^c \in \mathcal{L},$$

i.e. $B^c \in \mathcal{L}_A$.

(iii) Let $B_1, B_2, \cdots \in \mathcal{L}_A$ be disjoint. Then $A \cap B_n \in \mathcal{L}$ and these sets are disjoint, so

$$A \cap \bigcup_n B_n = \bigcup_n (A \cap B_n) \in \mathcal{L},$$

i.e. $\bigcup_n B_n \in \mathcal{L}_A$. □

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Dynkin's π-λ Theorem

Theorem (Dynkin)

Let $\mathcal{P}$ be a π-system and $\mathcal{L}$ a λ-system with $\mathcal{P} \subseteq \mathcal{L}$. Then

$$\sigma(\mathcal{P}) \subseteq \mathcal{L}.$$

Why care?

It is rarely interesting by itself, but it is used again and again: to prove that a property holds on a whole σ-field, check it on a generating π-system and show that the sets where the property holds form a λ-system.

Plan of the proof

Let $\mathcal{L}_0$ be the smallest λ-system containing $\mathcal{P}$. Show $\mathcal{L}_0$ is a π-system, hence a σ-field, hence $\sigma(\mathcal{P}) \subseteq \mathcal{L}_0 \subseteq \mathcal{L}$.

Dynkin's Theorem: Reduction

Step 0: Sandwich

Let $\mathcal{L}_0$ be the smallest λ-system containing $\mathcal{P}$. Since $\mathcal{L}_0$ is the intersection of all such λ-systems and $\mathcal{L}$ is one of them,

$$\mathcal{P} \subseteq \mathcal{L}_0 \subseteq \mathcal{L}.$$

What is left to show

If $\mathcal{L}_0$ is a σ-field, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}_0$ (a σ-field containing $\mathcal{P}$), hence $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.
By the previous lemma, $\mathcal{L}_0$ is a σ-field once it is a π-system. So it suffices to prove:

$$A, B \in \mathcal{L}_0 \implies A \cap B \in \mathcal{L}_0.$$

Dynkin's Theorem: Proof that $\mathcal{L}_0$ Is a π-System

Claim 1: for $C \in \mathcal{P}$, $\mathcal{P} \subseteq \mathcal{L}_{0,C}$

Let $C \in \mathcal{P}$; then $C \in \mathcal{L}_0$, so $\mathcal{L}_{0,C} = \{D : C \cap D \in \mathcal{L}_0\}$ is a λ-system (Lemma). If $D \in \mathcal{P}$, then $C \cap D \in \mathcal{P} \subseteq \mathcal{L}_0$ ($\mathcal{P}$ is a π-system), so $D \in \mathcal{L}_{0,C}$.

Claim 2: $C \in \mathcal{P},\ D \in \mathcal{L}_0 \Rightarrow C \cap D \in \mathcal{L}_0$

By Claim 1, $\mathcal{L}_{0,C}$ is a λ-system containing $\mathcal{P}$. By minimality, $\mathcal{L}_0 \subseteq \mathcal{L}_{0,C}$. So $D \in \mathcal{L}_0$ gives $C \cap D \in \mathcal{L}_0$.

Claim 3: for $C \in \mathcal{L}_0$, $\mathcal{P} \subseteq \mathcal{L}_{0,C}$

Let $D \in \mathcal{P}$. By Claim 2, $D \cap C \in \mathcal{L}_0$, so $D \in \mathcal{L}_{0,C}$.

Dynkin's Theorem: Conclusion

Claim 4 (the key claim)

Let $A \in \mathcal{L}_0$. By Claim 3, $\mathcal{P} \subseteq \mathcal{L}_{0,A}$, and $\mathcal{L}_{0,A}$ is a λ-system (Lemma). By minimality,

$$\mathcal{L}_0 \subseteq \mathcal{L}_{0,A}.$$

Now let $B \in \mathcal{L}_0$. Then $B \in \mathcal{L}_{0,A}$, i.e. $A \cap B \in \mathcal{L}_0$.

Finish

$\mathcal{L}_0$ is a π-system and a λ-system, hence a σ-field containing $\mathcal{P}$. Therefore

$$\sigma(\mathcal{P}) \subseteq \mathcal{L}_0 \subseteq \mathcal{L}. \qquad \square$$

Bookkeeping

The proof applies the $\mathcal{L}_A$ lemma three times (Claims 1, 2/3, 4), each time to push a condition from $\mathcal{P}$ up to all of $\mathcal{L}_0$ by minimality of $\mathcal{L}_0$.

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Main Theorem

Theorem

Let $\mathcal{A}_1, \ldots, \mathcal{A}_n \subseteq \mathcal{F}$ be **independent π-systems**. Then $\sigma(\mathcal{A}_1), \ldots, \sigma(\mathcal{A}_n)$ are independent.

Preparation

Put $\mathcal{B}_i = \mathcal{A}_i \cup \{\Omega\}$.

- $\mathcal{B}_i$ is still a π-system ($\Omega \cap A = A$).
- $\sigma(\mathcal{B}_i) = \sigma(\mathcal{A}_i)$, since any σ-field contains Ω.
- Independence of the $\mathcal{A}_i$ means $\mathbb{P}(B_1 \cap \cdots \cap B_n) = \prod_i \mathbb{P}(B_i)$ for all $B_i \in \mathcal{B}_i$.

Strategy

Replace one class at a time by its σ-field, using Dynkin's theorem at each stage.

Proof, Step 1: Replace $\mathcal{A}_1$ by $\sigma(\mathcal{A}_1)$

Define a class $\mathcal{L}$

Fix $B_2 \in \mathcal{B}_2, \ldots, B_n \in \mathcal{B}_n$ and put $B = B_2 \cap \cdots \cap B_n$. Let

$$\mathcal{L} = \Big\{ A \in \mathcal{F} : \ \mathbb{P}(A \cap B) = \mathbb{P}(A) \prod_{j=2}^{n} \mathbb{P}(B_j) \Big\}.$$

By independence, $\mathcal{B}_1 \subseteq \mathcal{L}$. We show that $\mathcal{L}$ is a λ-system.

(i) $\Omega \in \mathcal{L}$

$\mathbb{P}(\Omega \cap B) = \mathbb{P}(B_2 \cap \cdots \cap B_n) = \prod_{j \geq 2} \mathbb{P}(B_j) = \mathbb{P}(\Omega) \prod_{j \geq 2} \mathbb{P}(B_j)$, by independence of the $\mathcal{B}_j$.

(ii) Complements

Let $A \in \mathcal{L}$. By finite additivity, $\mathbb{P}(B) = \mathbb{P}(A \cap B) + \mathbb{P}(A^c \cap B)$, so

$$\mathbb{P}(A^c \cap B) = \prod_{j \geq 2} \mathbb{P}(B_j) - \mathbb{P}(A) \prod_{j \geq 2} \mathbb{P}(B_j) = \mathbb{P}(A^c) \prod_{j \geq 2} \mathbb{P}(B_j).$$

Proof, Step 1 (continued): (iii) and the Conclusion

(iii) Disjoint unions

Let $A_1, A_2, \cdots \in \mathcal{L}$ be disjoint. The sets $A_m \cap B$ are disjoint too, so by countable additivity

$$\mathbb{P}\Big(\bigcup_m A_m \cap B\Big) = \sum_m \mathbb{P}(A_m \cap B) = \sum_m \mathbb{P}(A_m) \prod_{j\geq 2} \mathbb{P}(B_j) = \mathbb{P}\Big(\bigcup_m A_m\Big) \prod_{j\geq 2} \mathbb{P}(B_j).$$

The product does not depend on m, so it is pulled out of the sum. Hence $\bigcup_m A_m \in \mathcal{L}$.

Apply Dynkin

$\mathcal{B}_1$ is a π-system, $\mathcal{L}$ is a λ-system and $\mathcal{B}_1 \subseteq \mathcal{L}$. So $\sigma(\mathcal{B}_1) = \sigma(\mathcal{A}_1) \subseteq \mathcal{L}$, i.e.

$$\mathbb{P}(A_1 \cap B_2 \cap \cdots \cap B_n) = \mathbb{P}(A_1) \prod_{j\geq 2} \mathbb{P}(B_j) \quad \text{for all } A_1 \in \sigma(\mathcal{A}_1),\ B_j \in \mathcal{B}_j.$$

Thus $\sigma(\mathcal{A}_1), \mathcal{A}_2, \ldots, \mathcal{A}_n$ are independent.

Proof, Step 2 and Induction

Replace $\mathcal{A}_2$ by $\sigma(\mathcal{A}_2)$

Fix $A_1 \in \sigma(\mathcal{A}_1)$ and $B_3 \in \mathcal{B}_3, \ldots, B_n \in \mathcal{B}_n$. Let

$$\mathcal{L} = \Big\{ A \in \mathcal{F} : \ \mathbb{P}(A_1 \cap A \cap B_3 \cap \cdots \cap B_n) = \mathbb{P}(A_1)\,\mathbb{P}(A) \prod_{j=3}^{n} \mathbb{P}(B_j) \Big\}.$$

Exactly as in Step 1, $\mathcal{L}$ is a λ-system; by Step 1, $\mathcal{B}_2 \subseteq \mathcal{L}$; and $\mathcal{B}_2$ is a π-system. Dynkin gives $\sigma(\mathcal{A}_2) \subseteq \mathcal{L}$.

Induction

Hence $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \mathcal{A}_3, \ldots, \mathcal{A}_n$ are independent. Repeating for $\mathcal{A}_3, \ldots, \mathcal{A}_n$ shows that $\sigma(\mathcal{A}_1), \ldots, \sigma(\mathcal{A}_n)$ are **mutually independent**. □

Extension to Infinitely Many Classes

Corollary

If $\{\mathcal{A}_i : i \in I\}$ is an independent family of π-systems (I countable or uncountable), then $\{\sigma(\mathcal{A}_i) : i \in I\}$ is an independent family.

Proof

By definition, independence of an infinite family means independence of every **finite** sub-family. Any finite sub-family $\{\sigma(\mathcal{A}_{i_1}), \ldots, \sigma(\mathcal{A}_{i_n})\}$ is independent by the Main Theorem applied to $\mathcal{A}_{i_1}, \ldots, \mathcal{A}_{i_n}$. □

Back to the counterexample

$\mathcal{A}_1 = \{A, B\}$ was **not** a π-system, since $A \cap B \notin \mathcal{A}_1$. That is exactly why the conclusion failed.

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Summary

Key results

Concept / Result	Content
Independent classes	product rule for $B_i \in \mathcal{A}_i \cup \{\Omega\}$
Counterexample	independent classes whose generated σ-fields are not independent
π-system	closed under finite intersections
λ-system	$\Omega \in \mathcal{L}$; closed under complements and countable disjoint unions
π & λ	a class that is both a σ-field
Dynkin's theorem	$\mathcal{P} \subseteq \mathcal{L} \Rightarrow \sigma(\mathcal{P}) \subseteq \mathcal{L}$
Main theorem	independent π-systems $\Rightarrow$ independent generated σ-fields

Exercises

1. Prove that the intersection of any family of λ-systems is a λ-system.
2. Prove that a λ-system $\mathcal{L}$ is a σ-field if and only if it is closed under finite intersections.
3. Show that if A and B are independent events, then $\sigma(\{A\}) = \{\varnothing, A, A^c, \Omega\}$ and $\sigma(\{B\})$ are independent σ-fields. Deduce that A and B^c are independent.
4. Complete the induction in the Main Theorem: write out the step from $\sigma(\mathcal{A}_1), \ldots, \sigma(\mathcal{A}_k), \mathcal{A}_{k+1}, \ldots, \mathcal{A}_n$ to $k+1$ replaced classes.

Outlook

Next lesson

The famed **Borel–Cantelli lemmas**: sufficient conditions for $\mathbb{P}(\limsup_n A_n)$ to be 0 or 1, where $\limsup_n A_n = \{A_n \text{ infinitely often}\}$.

Why this mattered

The second Borel–Cantelli lemma needs independence of the events, and the machinery of π-systems, λ-systems and Dynkin's theorem is what lets us manipulate independent families of events and σ-fields.